

22LPC402	LINEAR INTEGRATED CIRCUITS	L	T	P	C
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OBJECTIVES:

- To learn the linear and non-linear applications of operational amplifiers
- To learn the theory of ADC and DAC.
- To introduce the concepts of special function IC's.

UNIT I BASICS OF OPERATIONAL AMPLIFIERS 9

Ideal op-amp characteristics - General operational amplifier stages -and internal circuit diagrams of IC 741-Differential Amplifier-Common mode and differential mode gain, CMRR- op-amp DC characteristics - AC characteristics - Frequency response of op-amp - slew rate, Open and closed loop configurations.

UNIT II APPLICATIONS OF OPAMP 9

Inverting and Non-inverting amplifiers, differentiator, integrator, voltage to current converter, instrumentation amplifier, sine wave oscillator, low-pass, high pass and band-pass filters, comparator, multi-vibrators and Schmitt trigger, triangular wave generator, precision rectifier, log and antilog amplifiers.

UNIT III ANALOG TO DIGITAL AND DIGITAL TO ANALOG CONVERTERS 9

Analog switches, High speed sample and hold circuits, Types of D/A converter, Current driven DAC, Switches for DAC, A/D converter- Flash, Single slope, Dual slope, Successive approximation, Delta Sigma Modulation, ADC/DAC specifications.

UNIT IV TIMER AND PLL 9

Astable and monostablemultivibrators using IC 555 timer, applications of IC555 timer in astable mode and monostable mode, voltage-controlled oscillator, PLL -basic principle- analog and digital phase detector, PLL applications, frequency multiplication and translation, AM detector, FM demodulator, FSK modulators and demodulators, frequency synthesizers.

UNIT V SPECIAL FUNCTION ICS 9

Voltage regulators- IC voltage regulators, IC723 general purpose regulator, switching regulator, switched capacitor filter, voltage to frequency and frequency to voltage converters, power amplifier- monolithic power amplifier LM380.

TOTAL: 45 PERIODS

COURSE OUTCOMES

At the end of the course, the student will be able to

19/10/2025

Unit-1

Basics Of Operational Amplifier.

The electric device, active & passive components are fabricated on the single, chip semiconductor chip made of silicon material.

Advantage of Integrated circuits:

1. smaller in size.
2. It is highly reliable.
3. low cost.

4. low power consumption.

5. Low Noise.

[The main factor

we see power consumption, speed]

Amplifier:

It strengthens the weaker signal.

op-amp:

We can implement mathematical

function along with amplifications.

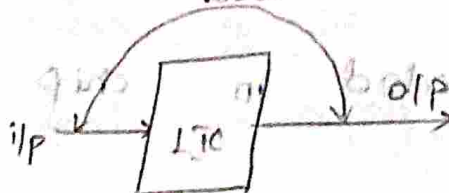
eg: it performs also operations like adding, integration, comparison etc...

LIC:

Relation btw input and output

of the particular LIC is linear

that's why it is linear Integrated circuits.



Classification of ICs based on types of signals.

- 1. Analog IC - process only analog signal.

eg: Regulated IC 741

- 2. Digital IC - process only digital sig

eg: AND, OR.

- 3. Mixed signal IC.

eg: ADC, DAC, PLL, etc

• Based on level of integration.

- 1. SSI - small scale integration.

⇒ 1 - 10 no. of input gates.
[First level of integration]

- 2. MSI - Medium scale integration

⇒ 10 - 100 no. of input gates can be integrated.

- 3. LSI - Large scale integration.

⇒ More than 1000 of inputs can

be integrated in chip.

- 4. VLSI - Very large scale integration.

⇒ millions of input gates can

be integrated in chip.

• Based on fabrication technology.

⇒ Thin film.

⇒ Thick film.

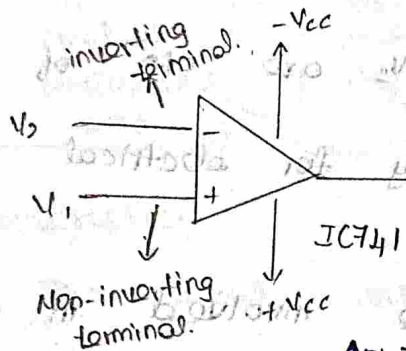
etc...

op-amp ⇒ IC $\mu A 741$.

↳ indicates the name of the company which involves manufacturing process.

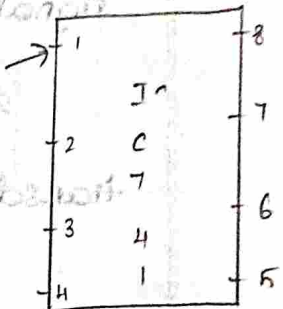
Internal blocks of operational amplifier:

Symbolic representation of IC.



$$V_0 = (A_{OL})(V_1 - V_2)$$

A_{OL} → open loop voltage gain of (DIP) op-amp



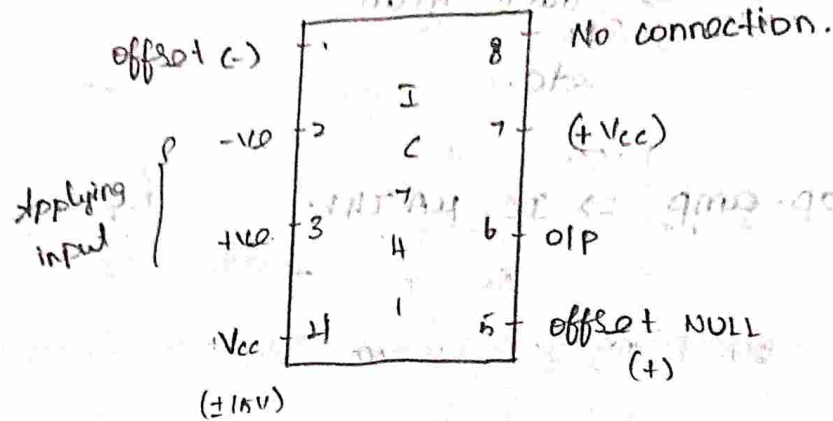
* If the IC which has pins on both sides is called DIP [Dual in line packaging].

* If the IC which has pins on single side of IC is called SIP [single in line packaging].

* If the IC which has pins on its four sides of IC is called QFP [quad flat packaging].

BAP [Blue array packaging] - pins on

TR0



supply voltage $\rightarrow \pm 12V$ to $\pm 15V$.
op-amp: It is used wherever the lowest signal to amplify.

Assume V_1 & V_2 are o/p of electrical transducer (Generally for electrical transducer is low)

$\Rightarrow$ 4 blocks are involved in op-amp.

* Differential Amplifier.

[Dual input and Dual ended o/p]
is called balanced differential amplifier.

$\Rightarrow$ The phase relation b/w i/p & o/p of transistor is 180° .

* If the input is given in the terminal it doesn't provide phase shift.

* If the i/p is given in the terminal of op-amp it provides 180° .

on

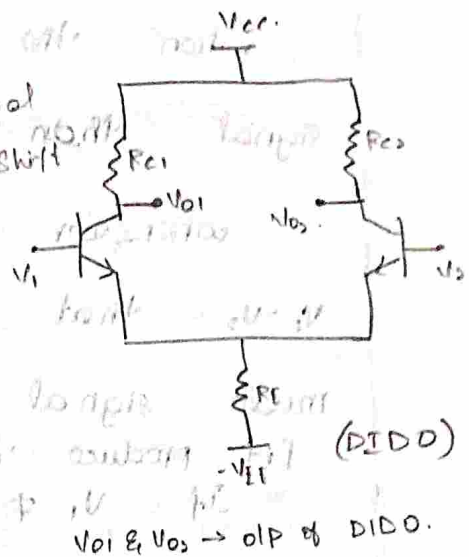
phase shift that's why is called

inverting terminal.

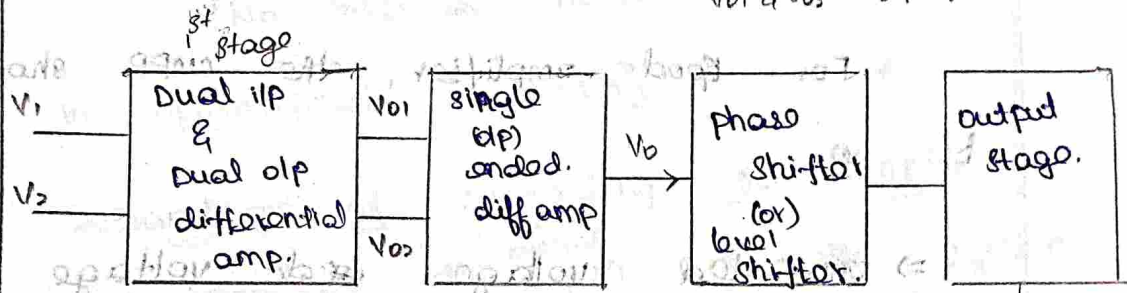
→ it amplifies weaker signal
by DDO with some phase shift
then use DDO next

phase shifter provide proper
180° phase shift the last 4th

is output stage (it has to
drain (DIDO)



V_{01} & $V_{02} \rightarrow$ o/p of DIDO.



characteristics:

The input impedance (R_i) of op-amp is correct (accurate).

① $\Rightarrow$ In ideal case, $R_i = \infty$ (In range 10^5 to $10^6 \Omega$)

dp impedance, $R_o = 0$

② $\Rightarrow$ In practical, $R_o < 100 \Omega$.

③ $\Rightarrow$ BW = ∞ (ideal case)
↳ indicates high frequency

It is range of frequency $\rightarrow$ BW.
for which the circuit exhibit good performance.

④ $\Rightarrow$ (Common mode Rejection Ratio) CMRR = ∞
CMRR = $\frac{A_d}{A_c} = \frac{\text{differential mode gain}}{\text{common mode gain}}$

* Gain is ratio of o/p to i/p.

* when the input is common mode signal then the gain $A_c = 0$

* whenever V_1 & V_2 are same (connected)
 $V_1 = V_2$ that signal is called common

mode signal, the o/p is A_c ($A_c = 0$)

[it produces amount of o/p]

* If $V_1 \neq V_2$, the gain circuit is A_d

* For good amplifier, the CMRR should be ∞

$\Rightarrow$ unwanted voltage or dc voltage

existing in input of operational

amplifier is known as offset voltage.

[presence of offset of V_{os} , output $\neq 0$] so.

By using offset (1 & 5 pin) we can avoid the offset voltage, to (cancel)

produce 'zero' o/p of the amplifier.

⑤ $\Rightarrow$ PSRR = ∞ [power supply rejection ratio]

The o/p of op-amp should not change even though some change in

power supply. (it should not affect o/p)

22/12/25

in order to provide stable output.

It is defined as ratio of change in input offset voltage (ΔV_{os}) to change in supply voltage (ΔV_s).

$$PSRR = \frac{\Delta V_{os}}{\Delta V_s} = \text{SVRR [Supply Voltage Rejection Ratio]}$$

(6). Zero Noise.

[No internal noise is produced in operational - amplifier].

Advantages of cascading the amplifier:

* Increase the overall gain of the system.

Internal blocks of op-amp.

1st stage. (DIDO op-amp)

* It ensures that input impedance

is very low.

* amplify the signal.

2nd stage.

* Voltage gain

(V_o) it's o/p is in usable form.

output stage.

* To ensure that the o/p impedance is low.

* Supply large amount of i's current

to drive the output circuit.

[op-amp amplifier cannot produce more than ± 15 V.

① If supply voltage is ± 15 V, saturated voltage of op-amp is ± 12 V.

If $V_{CC} = \pm 15$ V then saturated voltage of op-amp is ± 15 V.]

characteristics:

1) $R_{in} = \infty$ (practical $10^5 - 10^6 \Omega$)

2) $R_o = 0$ ($< 100 \Omega$).

3) $BW = \infty$.

4) $A_{OL} \Rightarrow$ open loop gain $= \infty$ ($> 10^6$)

5) $CMRR =$ Common Mode Rejection Ratio.

(For good amplifier,
(or) ideal)

$$CMRR = \frac{A_{dm}}{A_{cm}} = \infty$$

$\Rightarrow$ DIBO is also called symmetrical

differential amplifier.

(It's properties are same like

Resistors, Transistors)

purpose of D.D.O:

- It reject all common mode signal.

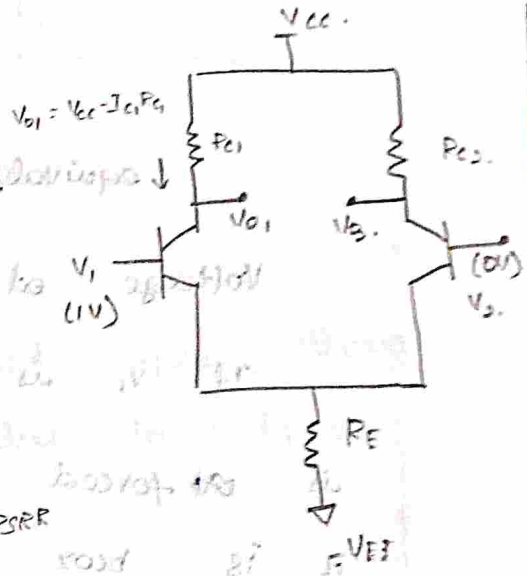
The differential amplifier responds only for different inputs of amplifier.

$$V_o = V_{o1} - V_{o2}$$

$$= V_{cc} - I_{c1} R_c - V_{cc}$$

$$V_o = -I_c R_c$$

some amount of o/p we get.



b) PSRR $\frac{\Delta V_{io}}{\Delta V_s} = \text{PSRR}$

$$\frac{\Delta V_o}{\Delta V_s}$$

PSRR & CMRR can be expressed in dB (or) normal.

7). zero noise.

8). zero drift.

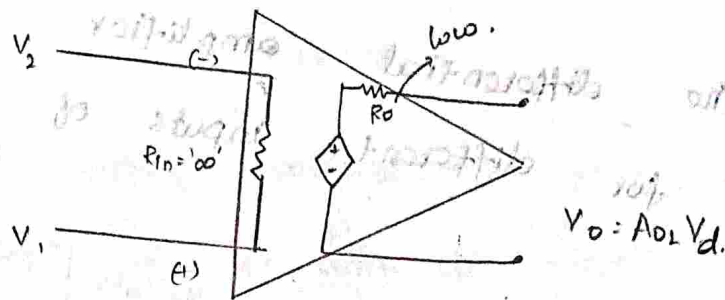
⇒ op-amp is not affected when

change in temperature.

9). zero i/p offset voltage. (in ideal case no i/p offset voltage).

10) slew Rate = '∞' (ideal) measured in terms of $\frac{V}{\mu s}$
Max. Rate of change of output w.r. to time.

Virtual Ground: (V_G)



equivalent circuit of IC 741.

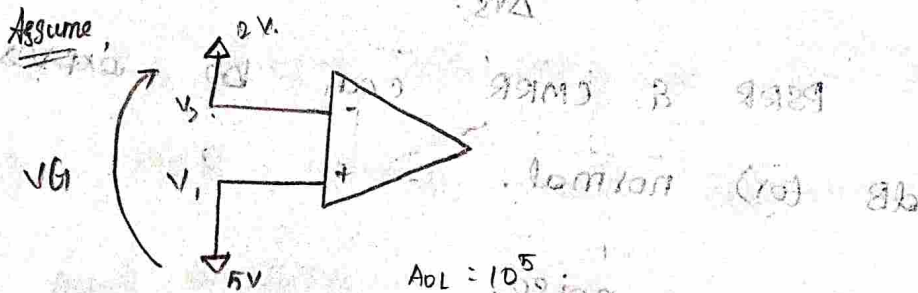
Voltage at V_1 & V_2 is same

If V_1 is given 0V then V_2

is forced to be 0V (indirectly)

It is because of virtual ground

of ideal op-amp



output of op-amp . $V_0 = A_{01}(V_1 - V_2)$

$$= A_{01}(5 - 2)$$

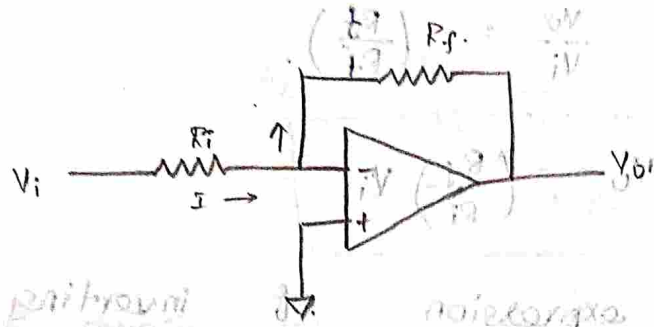
$$V_0 = 3 \times 10^5$$

since the o/p is more than +15V.

in this case the output of op-amp is saturated output +15V.

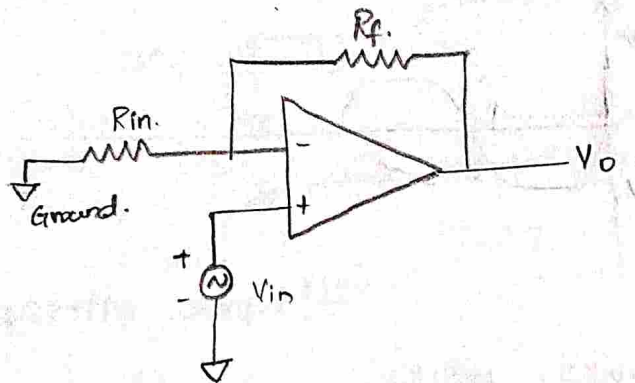
how quickly the output of op-amp responds to i/p (u) slow rate.
(or) measured by.

Inverting amplifier

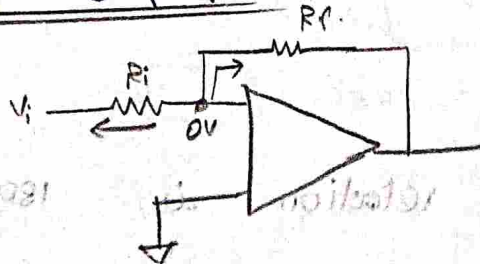


* There is no current flow through input of op-amp due to $R_i = \infty$

Non-Inverting amplifier



Inverting amplifier



apply Nodal analysis.

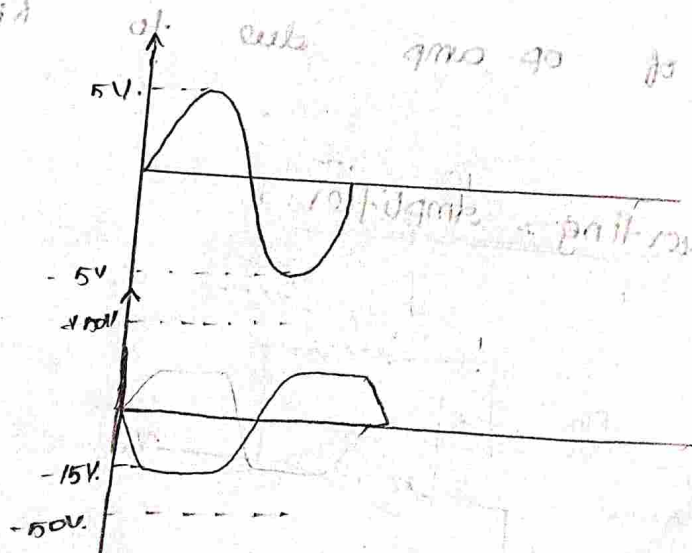
$$\frac{0 - V_i}{R_i} + \frac{0 - V_o}{R_f} = 0$$

$$\frac{-V_i}{R_i} = \frac{V_o}{R_f}$$

$$\frac{V_o}{V_i} = - \left(\frac{R_f}{R_i} \right)$$

$$V_o = - \left(\frac{R_f}{R_i} \right) V_i$$

output expression of inverting amplifier:-



if, $R_f = 10k\Omega$, $R_i = 1k\Omega$

$$V_o = - \left(\frac{10k}{1k} \right) V_i$$

$$= -10 \times 5 \quad \boxed{= -50V}$$

$\Rightarrow$ Phase relation is 180° phase shift.

$$A_{cl} = \frac{V_o}{V_i} = - \frac{R_f}{R_i}$$

$A_{cl} \rightarrow$ gain of closed loop.

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① Design an amplifier circuit to produce the gain of -5.

If gain is -ve, then it is inverting amplifier.

sol:-

Given, Gain = -5

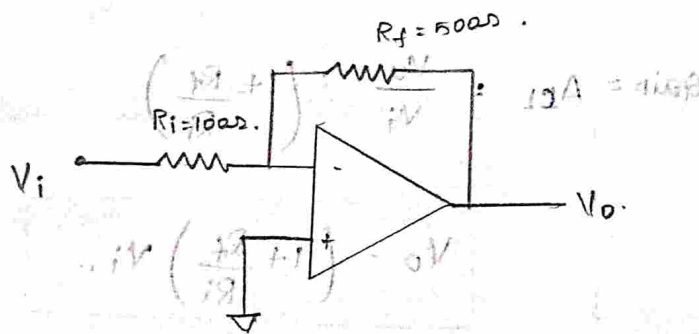
$$A_{cl} = -\frac{R_f}{R_i} = -5$$

$$R_f = 5 R_i$$

we choose, $R_i = 100\Omega$

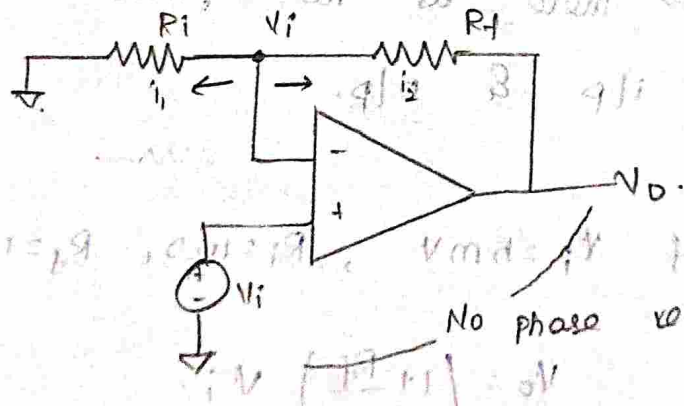
$$R_f = 5 \times 100\Omega$$

$$R_f = 500\Omega$$



Inverting Amplifier.

2. Non-inverting Amplifier.



No phase relationship.

By using Virtual Ground concept

Non-inverting terminal is V_i , so nodal

at inverting terminal is V_i
The current flows from higher potential to lower potential.

$$\frac{V_i - 0}{R_i} + \frac{V_i - V_o}{R_f} = 0.$$

$$\frac{V_i}{R_i} + \frac{V_i}{R_f} = \frac{V_o}{R_f}.$$

$$V_i \left[\frac{1}{R_i} + \frac{1}{R_f} \right] = \frac{V_o}{R_f}$$

$$\frac{V_o}{V_i} = \frac{R_i + R_f}{R_i} \times \frac{R_f}{R_f}.$$

$$\text{Gain} = A_{CL} = \frac{V_o}{V_i} = \left(1 + \frac{R_f}{R_i} \right)$$

$$V_o = \left(1 + \frac{R_f}{R_i} \right) V_i.$$

This is the output expression for Non-inverting amplifier.

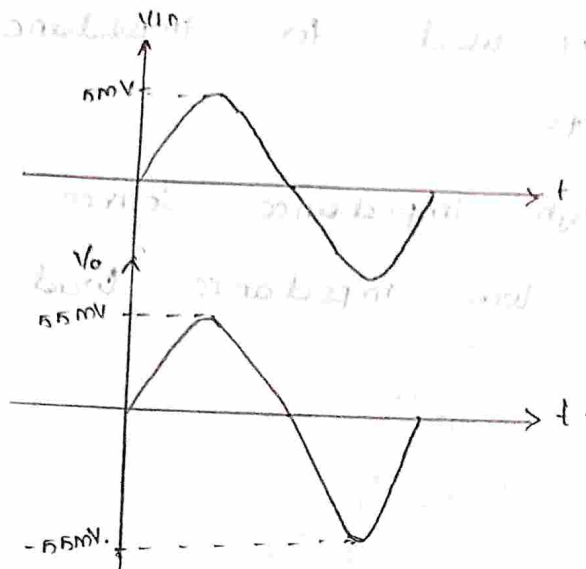
⇒ There is no phase relationship b/w i/p & o/p.

eg: if $V_i = 5 \text{ mV}$, $R_i = 10 \Omega$, $R_f = 100 \Omega$.

$$V_o = \left(1 + \frac{R_f}{R_i} \right) V_i.$$

$$= \left(1 + \frac{100}{10} \right) 5 \times 10^{-3}$$

$$V_o = 55 \text{ mV}.$$

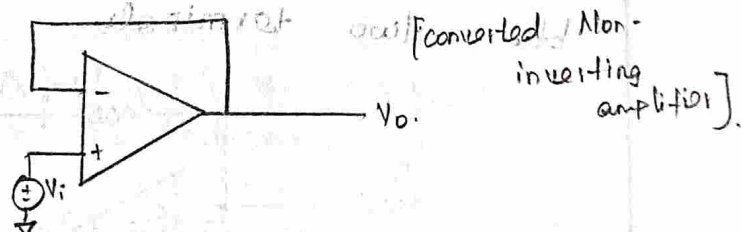


3. VOLTAGE FOLLOWER.

$$R_f = 0 \Omega.$$

$$R_i = \infty \Omega.$$

* The non-inverting amplifier is converted by assuming $R_f = 0 \Omega$, $R_i = \infty \Omega$



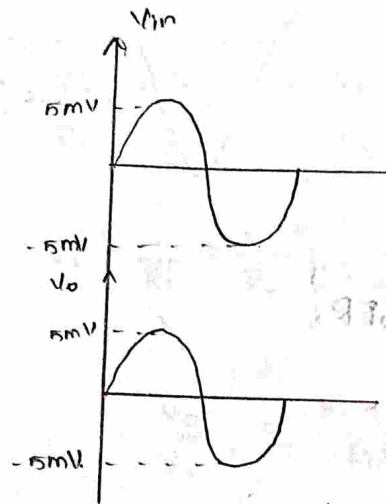
* The output simply follows the input ^{signal}, that's why it is called voltage follower.

$$\text{Gain} = \frac{V_o}{V_i} = 1 \quad [\because V_o = V_i]$$

* also known as unity gain amplifier circuit. (✓)

* It used for impedance matching circuits.

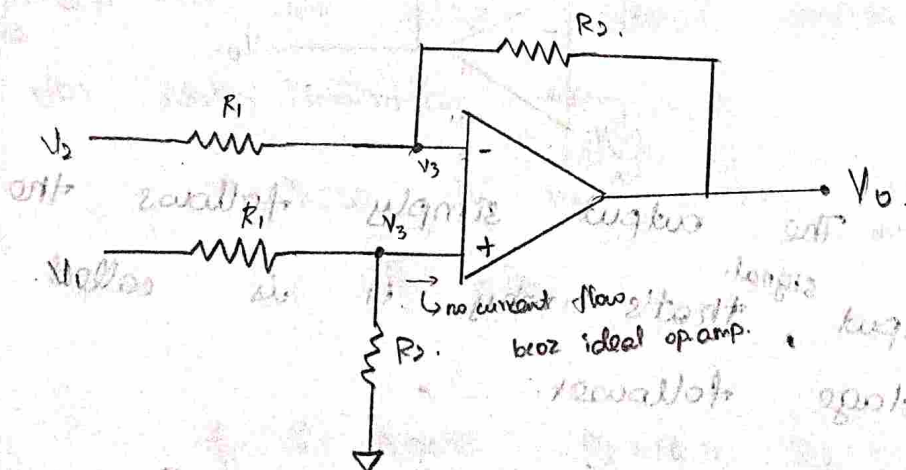
* High impedance source is connected with low impedance load $\Rightarrow$ impedance matching.



i/p & o/p waveforms are same.

DIFFERENTIAL AMPLIFIER / DIFFERENCE AMPLIFIER.

It will amplify the difference b/w two terminals.



Use virtual ground concept.

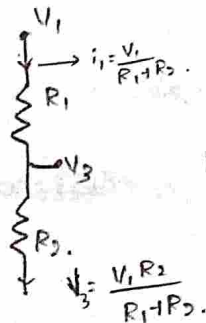
$$\frac{V_3 - V_1}{R_1} + \frac{V_3 - 0}{R_2} = 0$$

$$\frac{V_3}{R_1} + \frac{V_3}{R_2} = \frac{V_1}{R_1}$$

$$V_3 \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = \frac{V_1}{R_1}$$

$$V_3 = \frac{V_1}{\frac{R_1+R_2}{R_1 R_2} \cdot R_1}$$

$$V_3 = \frac{V_1 R_2}{R_1 + R_2} \rightarrow \textcircled{1}$$



\$R_1\$ & \$R_2\$ series.

apply nodal analysis w.r.t \$(-V_2)\$

terminal of op-amp \$o_1\$:

$$\frac{V_3 - V_2}{R_1} + \frac{V_3 - V_0}{R_2} = 0$$

$$V_3 \left[\frac{1}{R_1} + \frac{1}{R_2} \right] - \frac{V_2}{R_1} = \frac{V_0}{R_2}$$

$$V_3 \left[\frac{R_1 + R_2}{R_1 R_2} \right] - \frac{V_2}{R_1} = \frac{V_0}{R_2} \rightarrow \textcircled{2}$$

sub \$N_3\$,

$$\frac{V_1 R_2}{R_1 + R_2} \left[\frac{R_1 + R_2}{R_1 R_2} \right] - \frac{V_2}{R_1} = \frac{V_0}{R_2}$$

$$\frac{V_1}{R_1} - \frac{V_2}{R_1} = \frac{V_0}{R_2}$$

$$\frac{1}{R_1} [V_1 - V_2] = \frac{V_0}{R_2}$$

$$V_o = \frac{R_2}{R_1} [V_1 - V_2]$$

* The output expression is amplified version of difference btw V_1 & V_2 .

eg:

$$V_1 = 3\text{mV}, V_2 = 2\text{mV}$$

$$R_1 = 10\text{k}\Omega, R_2 = 1000\text{k}\Omega$$

$$V_o = \frac{R_2}{R_1} (V_1 - V_2)$$

$$= \frac{1000\text{k}}{10\text{k}} (3\text{m} - 2\text{m})$$

$$= 100 (1\text{m}) \text{ V}$$

$$V_o = 0.1 \text{ V}$$

* Instrumentation amplifier means amplifies the weak (or) low voltage signal.

* It is used in instrumentation Amplifier.

(uses of Difference amplifier 2m)

* High input impedance & low output impedance.

For amplifier circuit, the transistor should be in the linear mode for that biasing method is used.

$V_d = (V_1 - V_2)$ difference b/w two terminal.
 $V_{cm} = \frac{V_1 + V_2}{2}$ average of two terminal.

$$V_o = A_1 V_1 + A_2 V_2 \rightarrow (1)$$

consider, $A_1 = \frac{V_o}{V_1}$ at $V_2 = 0$.

$A_2 = \frac{V_o}{V_2}$ at $V_1 = 0$.

$$V_1 = V_{cm} + \frac{V_d}{2}$$

$$V_2 = V_{cm} - \frac{V_d}{2}$$

sub eq (2) in (1)

$$V_o = A_1 \left[V_{cm} + \frac{V_d}{2} \right] + A_2 \left[V_{cm} - \frac{V_d}{2} \right] \quad (1)$$

$$V_o = V_{cm} [A_1 + A_2] + \frac{V_d}{2} [A_1 - A_2]$$

$$V_o = A_{cm} V_{cm} + A_{dm} V_d \rightarrow (3)$$

Compare eq (1) & (3)

$$A_{cm} = A_1 + A_2$$

$$A_{dm} = \frac{1}{2} (A_1 - A_2)$$

$$\delta = \text{CMRR} = \left| \frac{A_{om}}{A_{cm}} \right| \rightarrow \text{ideal op-amp} = \infty$$

$$\rightarrow \text{Practical op-amp} \approx 70 \text{ dB}$$

DC CHARACTERISTICS OF OP-AMP:

$\Rightarrow$ whenever the op-amp is non-ideal, it is defined as DC & AC characteristics.

$\Rightarrow$ Non-ideal op-amp — $\begin{cases} \rightarrow \text{dc characteristics.} \\ \rightarrow \text{AC characteristics.} \end{cases}$

$\Rightarrow$ 4 DC characteristics -

- 1). Input Bias current.
- 2). Input offset current.
- 3). Input offset voltage.
- 4). Thermal drift.

①. Input Bias Current.

I_B^- current enters to the inverting terminal.

There must be small amount of current flow through the base of the transistor. There must be some imbalance so in practical I_{B^+}, I_{B^-} it is not same.



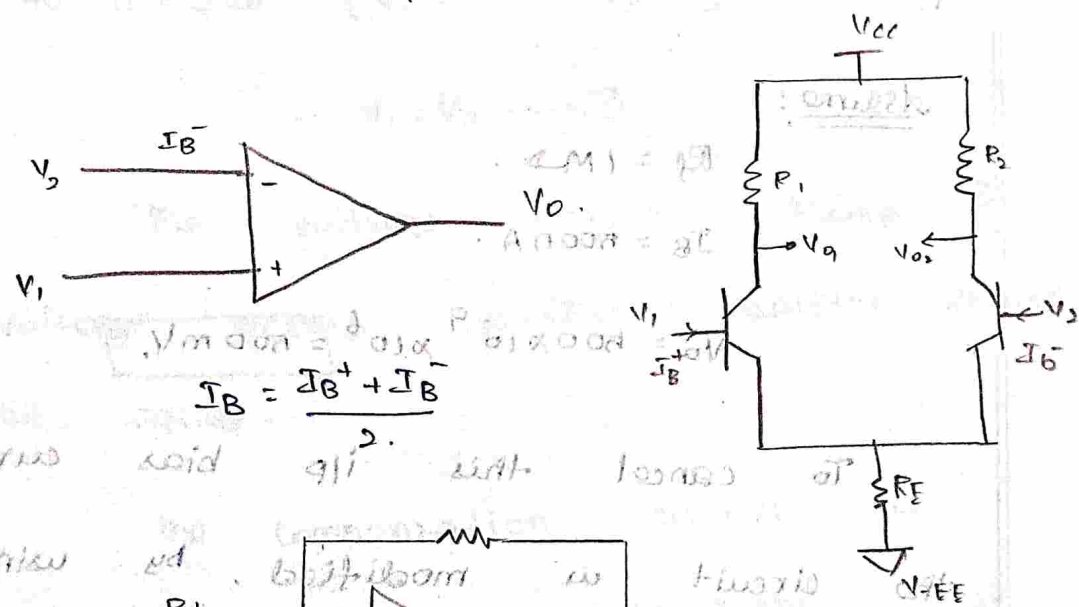
($V_1 = 0$)

output

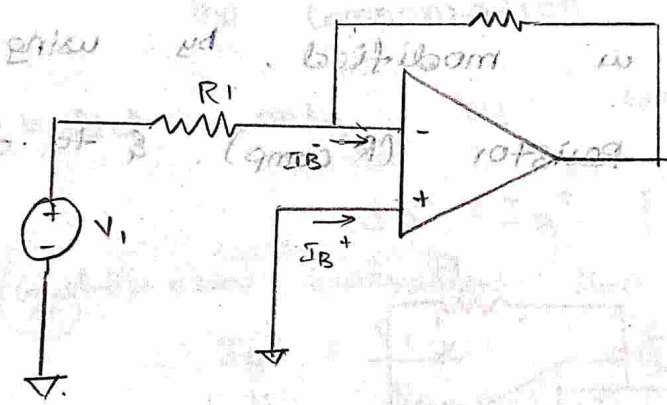
∞
 $\approx 70 \text{ dB}$
 no current
 ideal.
 characteristic

$I_B \rightarrow$ average of current flowing
 through two terminal of op-amp
 is known as input bias current.

For IC-741, practical $I_B = 500 \text{ nA}$.

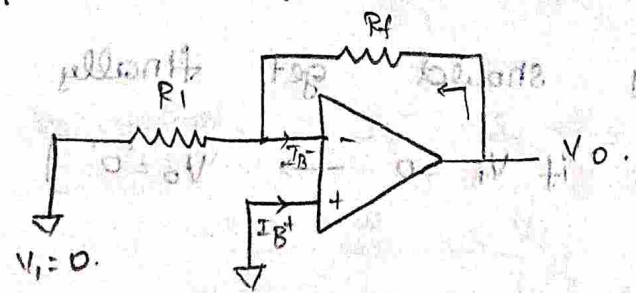


$$I_B = \frac{I_{B^+} + I_{B^-}}{2}$$



($V_1 = 0 \rightarrow V_o = 0$).

Due to some bias current the
 output voltage is not zero.



$$V_o = (I_{B^-} R_f)$$

Even though the input is zero the output has some value instead of getting 'zero' it bcoz due to i/p bias current.

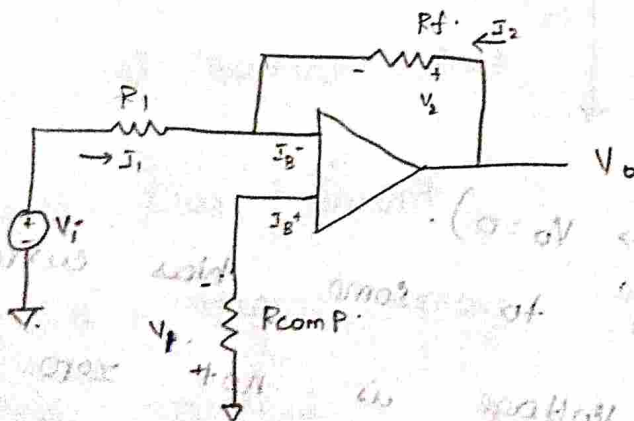
Assume:

$$R_f = 1 \text{ M}\Omega$$

$$I_B = 500 \text{ nA}$$

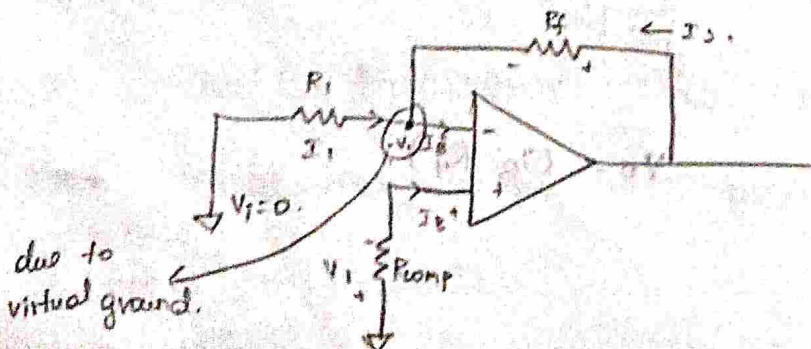
$$V_o = 500 \times 10^{-9} \times 10^6 = 500 \text{ mV}$$

To cancel this i/p bias current, the circuit is modified, by using compensation Resistor (R_{comp}). & to find R_{comp} .



[the circuit should get finally

$$\text{if } V_i = 0 \rightarrow V_o = '0']$$



Apply KVL,

$$V_0 - V_2 - (-V_1) = 0$$

$$V_0 = V_2 - V_1 \rightarrow (1)$$

to achieve $[V_1 = 0 \rightarrow V_0 = 0]$.

$$\therefore V_1 = V_2 \rightarrow (2)$$

The voltage across R_{comp} & voltage across feedback resistor should be equal.

The compensation circuit is

valid only by considering that

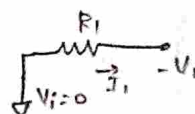
$$I_B^- = I_B^+ \text{ [circuits work only]}$$

(condition for bias current flows through two terminal.)

$$I_B^+ = \frac{V_1}{R_{comp}} \rightarrow (3)$$

$$I_1 = \frac{V_1 - (-V_1)}{R_1} = \frac{0 + V_1}{R_1}$$

$$I_1 = \frac{V_1}{R_1} \rightarrow (4)$$



by nodal, $I_B^- = I_1 + I_2$



$$I_B^- = \frac{V_1}{R_1} + \frac{V_2}{R_f} \rightarrow (5)$$

equating equation (3) & (5).

$$\frac{V_1}{R_{comp}} = \frac{V_1}{R_1} + \frac{V_2}{R_f} \quad [\because V_1 = V_2]$$

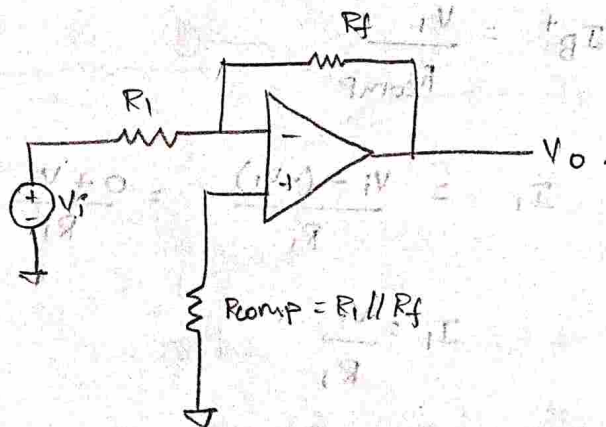
$$\frac{V_1}{R_{comp}} = \frac{V_1}{R_1} + \frac{V_1}{R_f}$$

$$\frac{V_1}{R_{comp}} = V_1 \left[\frac{1}{R_1} + \frac{1}{R_f} \right]$$

$$\frac{1}{R_{comp}} = \frac{1}{R_1} + \frac{1}{R_f}$$

$$\therefore R_{comp} = \frac{R_1 R_f}{R_1 + R_f} = R_1 \parallel R_f$$

to compensate the effect due to i/p bias current, the $(R_{comp} = R_1 \parallel R_f)$ is connected in non-inverting terminal.



②. Input offset current. (I_{os})

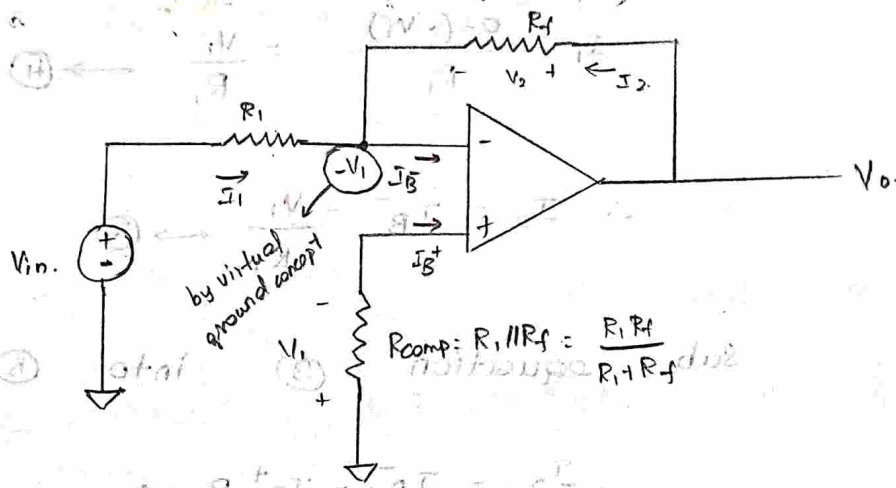
difference between

two terminal current of op-amp

Generally,

$$|I_{os}| = I_B^+ - I_B^- = I_B^- - I_B^+$$

In practical value of $I_{os} = 200 \text{ nA}$



If two terminal current is not

same then we use this.

$$[KVL] \quad V_o - V_2 - (-V_1) = 0$$

$$V_o - V_2 + V_1 = 0$$

$$V_o = V_2 - V_1$$

$$V_2 = I_2 R_f$$

$$V_1 = I_B^+ R_{comp}$$

I_B^- & I_B^+ are not equal

if so what is equation we find.

write KCL equation w.r.t (-ve)

terminal of op-amp.

$$I_1 + I_2 = I_B^-$$

$$I_2 = I_B^- - I_1$$

Assume, $V_i = 0$.

$$I_1 = \frac{0 - (-V_i)}{R_1} = \frac{V_i}{R_1} \rightarrow (4)$$

$$\therefore I_2 = I_B^- - \frac{V_i}{R_1} \rightarrow (5)$$

Sub equation (3) into (5).

$$I_2 = I_B^- - \frac{I_B^+ R_{comp}}{R_1} \rightarrow (6)$$

Apply equation (2), (3), (6) into equation (1).

$$V_o = I_2 R_f - I_B^+ R_{comp}$$

$$V_o = \left[I_B^- - \frac{I_B^+ R_{comp}}{R_1} \right] R_f - I_B^+ R_{comp}$$

$$V_o = I_B^- R_f - I_B^+ R_{comp} \left[\frac{R_f}{R_1} + 1 \right]$$

$$\therefore V_o = I_B^- R_f - I_B^+ R_{comp} \left[\frac{R_1 + R_f}{R_1} \right] \cdot \frac{R_f}{R_f}$$

$$V_o = I_B^- R_f - I_B^+ R_{comp} \left[\frac{R_1 + R_f}{R_1 R_f} \right] R_f$$

$$V_o = I_B^- R_f - I_B^+ R_f$$

$$V_o = (I_B^- - I_B^+) R_f$$

$$V_o = |I_{os}| R_f$$

For IC741, $|I_{os}| = 500 \text{ nA}$ assume, $R_f = 1 \text{ M}\Omega$.

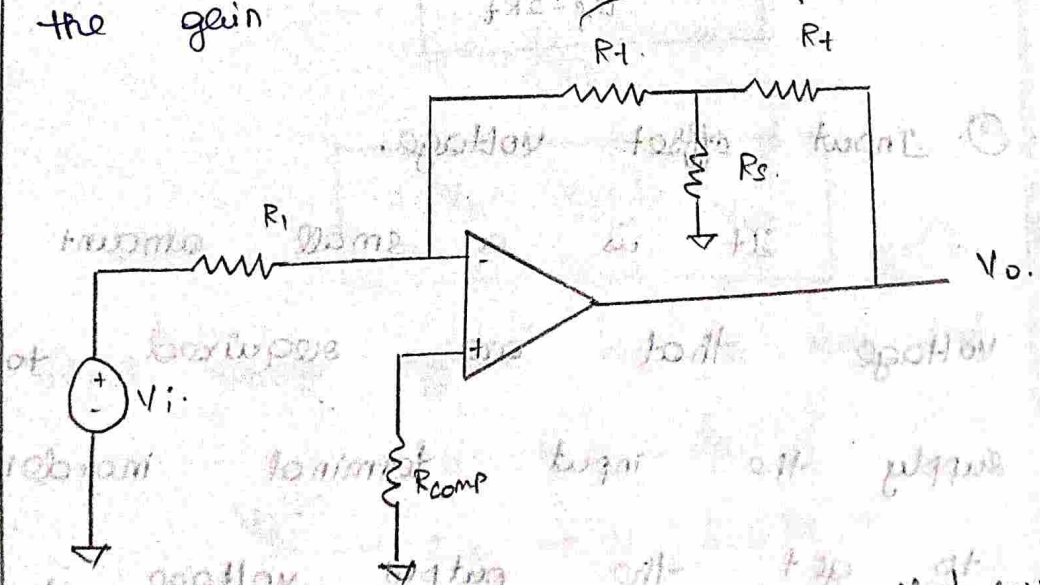
$$V_o = 500 \times 10^{-9} \times 1 \times 10^6$$

$$= 500 \times 10^{-3}$$

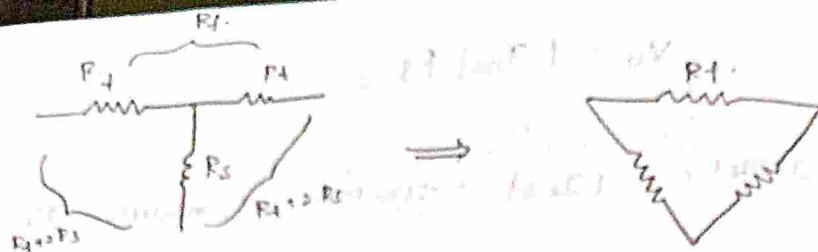
$$V_o = 500 \text{ mV.}$$

When no i/p is applied but V_o is getting 500 mV because of i/p offset current.

if $|I_{os}| \uparrow$, $V_o \uparrow$ so we need to cancel this effect.
we need to choose proper R_f value to reduce this effect.
at the same time it should not affect the gain



modified circuit to cancel i/p offset current at the same time not affect the gain.



[T-network]

$$R_{AB} = \frac{R_A + R_B + R_C}{R_C}$$

$$R_f = R_t + R_t + \frac{R_t R_t}{R_s}$$

$$R_f = \frac{2R_t R_s + R_t^2}{R_s}$$

choose, $R_t \ll R_f/2$

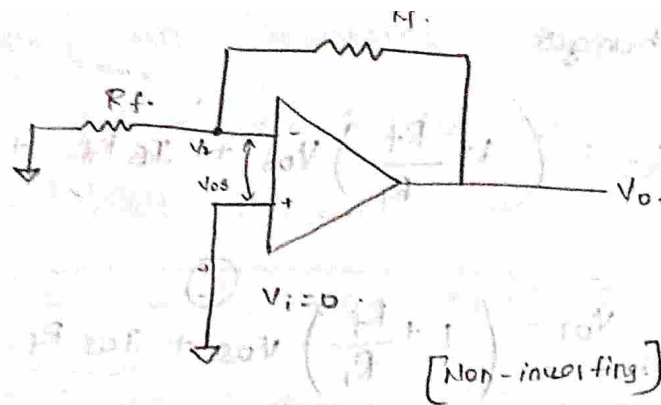
$$2R_t + \frac{R_t^2}{R_s} = R_f$$

$$\frac{1}{R_s} = \frac{R_f - 2R_t}{R_t^2}$$

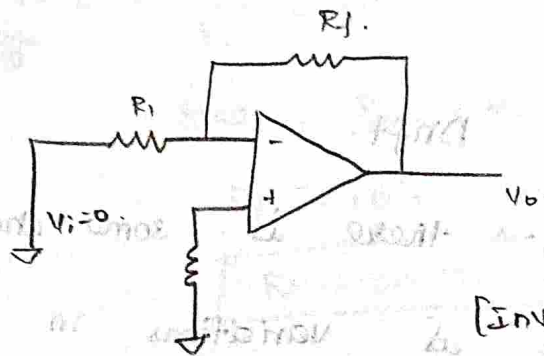
$$R_s = \frac{R_t^2}{R_f - 2R_t}$$

③. Input offset voltage.

It is a small amount of voltage that are required to supply the input terminal in order to get the output voltage is zero (when V_1 & $V_2 = 0$).



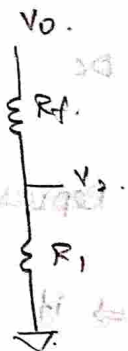
$V_i = 0 \Rightarrow V_i = 0$
 $V_i = 0$ is shorted



voltage divided rule.

$$V_2 = \frac{V_o}{R_1 + R_f} \cdot R_1$$

$$V_{os} = V_i - V_2$$



$$\therefore V_i = 0$$

$$|V_{os}| = V_2$$

$$V_{os} = \frac{V_o}{R_1 + R_f} \cdot R_1$$

$$\therefore V_o = \frac{R_1 + R_f}{R_1} \cdot V_{os}$$

output voltage affected by for non-ideal op-amp.

$$I_B \rightarrow V_o = I_B R_f$$

$$I_{os} \rightarrow V_o = I_{os} R_f$$

$$V_{os} \rightarrow V_o = \left(1 + \frac{R_f}{R_1}\right) V_{os}$$

Total changes.

[When R_{comp} is not known]

$$V_{OT} = \left(1 + \frac{R_f}{R_i} \right) V_{OS} + \underbrace{I_B R_f}_{\text{compensated by using } R_{comp}} + I_{AS} R_f.$$

neglected.

[R_f is known & designed]

$$V_{OT} = \left(1 + \frac{R_f}{R_i} \right) V_{OS} + \underbrace{I_{AS} R_f}_{\substack{\text{Vp offset current} \\ \text{affects strongly}}}.$$

④. Thermal Drift.

drift $\rightarrow$ there is some changes.

* There are variations in 8 other DC characteristics depends on temperature.

Representation (or) expressed, in terms.

$\Rightarrow$ if drift happens on:

$$\left. \begin{array}{l} \Delta I_P \text{ bias current} \\ \Delta I_P \text{ offset current} \end{array} \right\} \rightarrow \text{mA}/^{\circ}\text{C}$$

$\Rightarrow$ if drift happens on:

$$\Delta I_P \text{ offset voltage} \rightarrow \text{mV}/^{\circ}\text{C}$$

we can prevent the thermal drift

* By proper PCB design!

* using cooling circuits.

① Design the using of solt.

① Design an inverting amplifier to compensate the effect due to input offset current using IC741 op-amp to get a gain of -10 and input impedance of $10\text{ M}\Omega$.

Soln.

$$A_{cl} = \frac{-R_f}{R_i} = -10.$$

Given, $R_i = 10\text{ M}\Omega$.

$$R_f = 10 \times 10\text{ M}\Omega.$$

$$R_f = 100\text{ M}\Omega.$$

$$R_f \ll R_{f/2}$$

$$R_f \ll \frac{100}{2}\text{ M}\Omega.$$

$$R_f \ll 50\text{ M}\Omega.$$

choose

$$R_f = 45\text{ M}\Omega.$$

$$R_s = \frac{R_f^2}{R_f - 2R_f}$$

$$= \frac{(45 \times 10^6)^2}{100 \times 10^6 - 2 \times 45 \times 10^6}$$

$$= \frac{2025 \times 10^{12}}{10^6 (100 - 90)}$$

$$= \frac{2025}{10} \times 10^6$$

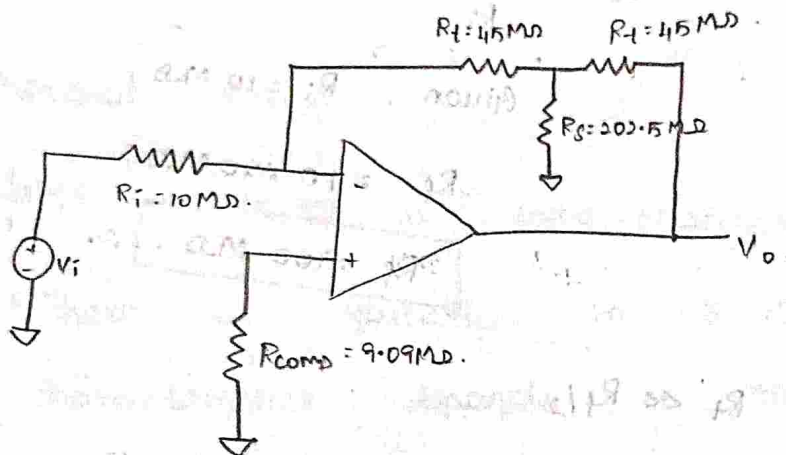
$$R_s = 202.5\text{ M}\Omega$$

$$R_{comp} = R_1 \parallel R_f$$

$$= \frac{R_1 R_f}{R_1 + R_f} = \frac{10 \times 10^6 \times 100 \times 10^6}{110 \times 10^6}$$

$$= \frac{1000}{110} \times 10^6$$

$$R_{comp} = 9.09 \text{ M}\Omega$$



AC CHARACTERISTICS OF AN OPERATIONAL AMPLIFIER:

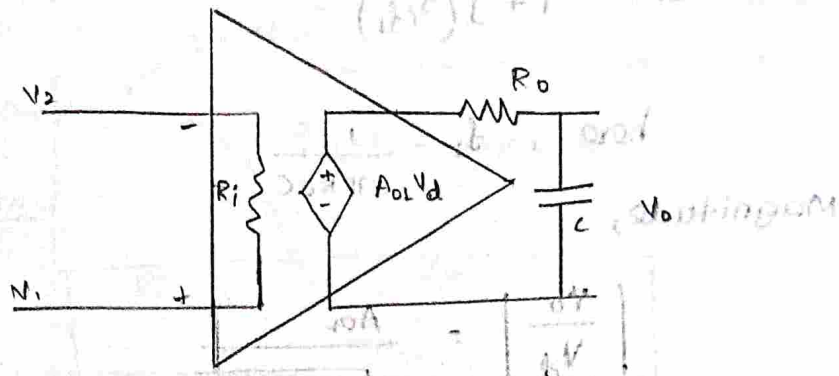
- 1). Frequency Response
- 2). Slew Rate
- 3). Stability

[What about the output of system when the frequency changes - frequency Response.]

①. Frequency Response:

open loop

$$V_o = A_{OL} (V_1 - V_2)$$



$C \rightarrow$ due to parasitic capacitor existing in the internal op-amp.

$R_o \rightarrow$ output Resistance (ideal $R_o = 0$)

$R_i \rightarrow$ input Resistance (for ideal $R_i = \infty$)

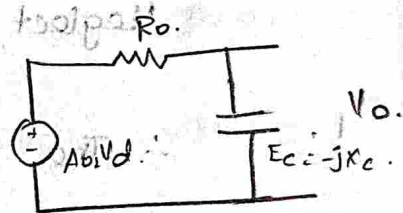
$A_{ol} \rightarrow$ open loop gain.

$V_d \rightarrow$ difference btw

terminals voltage.

By voltage divided,

$$V_o = \left[\frac{A_{ol} V_d}{R_o - j\omega C} \right] (-j\omega C)$$



$$V_o = \frac{A_{ol} V_d}{-j\omega C [1 + j\omega R_o C]}$$

$$V_o = \frac{A_{ol} V_d}{1 + j\omega R_o C}$$

$$\frac{V_o}{V_d} = \frac{A_{ol}}{1 + j\omega R_o C}$$

$$\frac{V_o}{V_d} = \frac{A_{ol}}{1 + j2\pi f R_o C}$$

$$V_o = \frac{R_o}{R_i + R_o} \cdot V_i$$

$\downarrow$ $\downarrow$ $\downarrow$
 V_o R_o $A_{ol} V_d$

$$\frac{V_o}{V_d} = \frac{A_{OL}}{1 + j(f/f_L)}$$

$f/f_L \rightarrow$ frequency due
to o/p resistance &
i/p resistance

here, $f_L = \frac{1}{2\pi R_{OL} C}$

Magnitude,

$$\left| \frac{V_o}{V_d} \right| = \frac{A_{OL}}{\sqrt{1 + (f/f_L)^2}}$$

$f_L \rightarrow$ can't change.
 $f \rightarrow$ can vary.

Cases:

i). $f = f_L \Rightarrow \left| \frac{V_o}{V_d} \right| = \frac{A_{OL}}{\sqrt{2}}$

ii). $f \ll f_L \Rightarrow f/f_L \ll 1$

* Neglect the term f/f_L compare to

1. $\therefore$ The gain is constant. (same as A_{OL})

iii). $f \gg f_L \Rightarrow f/f_L \gg 1$

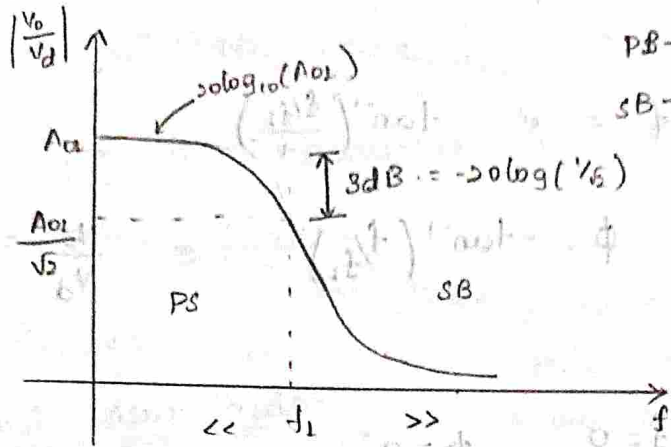
* Neglect 1 compared to f/f_L

$$\left| \frac{V_o}{V_d} \right| = \frac{A_{OL}}{\sqrt{(f/f_L)^2}}$$

$$\left| \frac{V_o}{V_d} \right| = \frac{A_{OL} f_L}{f}$$

f increases then the gain $\frac{V_o}{V_d}$ decreases.

Magnitude Response:



f_1 : $\Rightarrow$ 3 db frequency - frequency at which gain is reduced to.

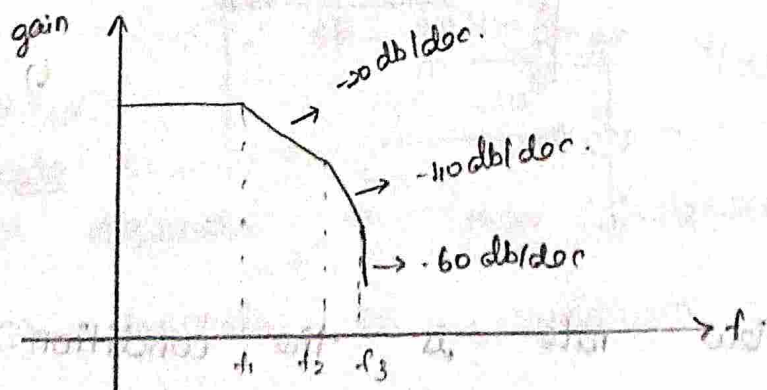
70.7% of its Max gain
 0.707 Max gain.

PB $\rightarrow$ Range of frequency which allow all the frequency. (0 to 10 kHz).

SB $\rightarrow$ from (10 kHz) it attenuates.

#. for order of system is 2 and then -20 dB/dec .

[Assume n no. of op. amp is cascaded.]



to find phase Response of system.

→ No real part in Numerator.

$$\phi = 0^\circ - \tan^{-1}\left(\frac{f}{f_L}\right)$$

$$\phi = -\tan^{-1}\left(\frac{f}{f_L}\right)$$

$$\frac{V_o}{V_d} = \frac{A_{OL}}{1+j(f/f_L)}$$

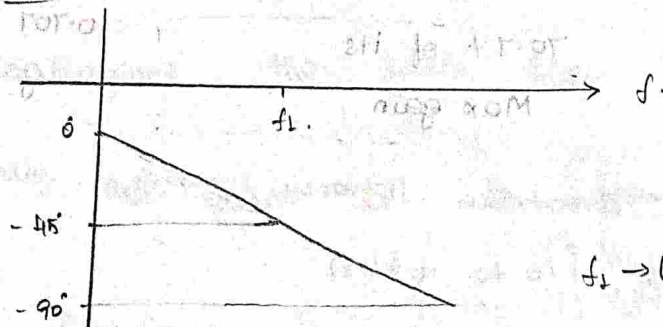
cases:

$$f = 0, \phi = 0^\circ$$

$$f = f_L, \phi = -45^\circ$$

$$f = \infty, \phi = -90^\circ$$

Phase Response:



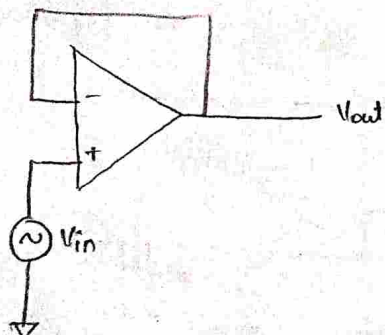
$f_L \rightarrow$ lies btw 0° to 45°

$\Rightarrow$ max phase shift

is -90° for RC circuit.

②. Slew Rate: (SR)

or



$$V_{in} = V_{out}$$

Slew rate is the condition for which we get distortion less as output for input applied.

Max rate of change output w.r. to time for the applied input

It is measured in $V/\mu s$ unit.

$$SR = \left. \frac{dV_{out}(t)}{dt} \right|_{\max}$$

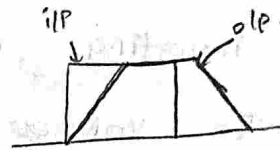
$\Rightarrow$ when capacitor is not there no sudden change in raise or fall, it changes gradually.

to find slew rate:

$$Q = C V_{out}$$

$$\frac{dQ}{dt} = C \frac{dV_{out}}{dt}$$

$$I = C \frac{dV_{out}}{dt}$$



op is not same as ip due to presence of capacitance affect inside the op-amp.

$\therefore$ for ideal case 'SR' must be " ∞ "

$$SR \Rightarrow \boxed{\frac{dV_{out}}{dt} = \frac{I}{C}}$$

for voltage follower circuit.

$$V_{out} = V_{in} = V_m \sin \omega t$$

$$\frac{dV_{out}}{dt} = V_m \cos \omega t (\omega) = V_m \omega \cos \omega t$$

$$\left. \frac{dV_{out}}{dt} \right|_{\max} = V_m \omega = 2\pi V_m f_{\max} = SR$$

$$\therefore SR = \frac{2\pi V_m f_{\max}}{10^6} \text{ V}/\mu s$$

Slow Rate:

* It depends on max frequency of input ($f_{\max}$)

* It depends on maximum amplitude

of input signal (V_m)

f_{max} - Maximum frequency of the input in order to get the distortion less output.

⇒ For ICTH1, the slew rate is $0.5 \text{ V}/\mu\text{sec}$

[SR should express only in $\text{V}/\mu\text{sec}$]

⑥. A ICTH1 op-amp is used as an inverting amplifier with gain of 50.

The voltage gain vs frequency curve of ICTH1 is flat upto 20 kHz , the slew rate of ICTH1 is $0.5 \text{ V}/\mu\text{sec}$ what maximum peak to peak input signal can be applied without distorting the output.

Sol:

Given, $f_{max} = 20 \text{ kHz}$

Gain = 50.

SR = $0.5 \text{ V}/\mu\text{sec}$.

$$S.R = \frac{2\pi V_m f_{max}}{10^6} \text{ V}/\mu\text{s}$$

$$0.5 = \frac{2 \times \pi \times V_m \times 20 \times 10^3}{10^6}$$

$$0.5 \times 10^3$$

$$\frac{2 \times \pi \times 20}{10^6} = V_m$$

The applied

$$V_m = \frac{500}{40 \times \pi}$$

$$V_m = \frac{500}{125.6637}$$

$$V_m = 3.9788 \text{ V}$$

$$V_{p-p} = 2 V_m = 2 \times 3.9788 \text{ V.}$$

$$V_{p-p} = 7.9577 \text{ V.}$$

$$A = \frac{V_o}{V_{in}}$$

$$V_o = 50 \times 3.9788$$

$$V_o = 198.9 \text{ V.}$$

$$(V_{in} = V_m)$$

$$V_o = 50 \times 7.96$$

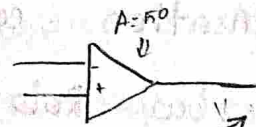
$$V_o = 398 \text{ V.}$$

$$(V_{in} = V_{p-p})$$

0.5 V/μsec ← SR

$$V_m = 3.9788$$

$$V_{p-p} = 7.9577 \text{ V.}$$



for this what $V_{in} = ?$

$$V_o = A_v V_{in}$$

$$V_{in} = \frac{7.96}{50}$$

$$V_{in} = 0.159 \text{ V}$$

$$V_{in} = 159 \text{ mV.}$$

The ilp signal of 159 mV can be applied without distorting the output.

③. A non-inverting amplifier with gain of ∞ is null at 25°C . what will happen to the output voltage if temperature rises to 50°C for an offset voltage drift of $0.15\text{ mV}/^\circ\text{C}$.

Sol:-

Given, $A = 100$.

offset voltage drift $0.15\text{ mV}/^\circ\text{C}$.

$V_o = A (\Delta T) \times \text{drift per } 1^\circ\text{C}$

$V_o = 100 \times 25 \times 0.15 \times 10^{-3}$

$V_o = 375\text{ mV}$

④. If the bias current in the JFET op-amp is $19\text{ }\mu\text{A}$ & the internal frequency compensation capacitor is 30 pF then find the slew rate of op-amp.

$S.R = \frac{I}{C}$

$= \frac{19 \times 10^{-6}}{30 \times 10^{-12}}$

$= \frac{0.633}{10^{-6}}$

$S.R = 0.633\text{ V}/\mu\text{s}$

Rise time for op-amp:

$$t_r = \frac{0.35}{BW}$$

time taken to output to reach its 10% to 90% of its maximum value.

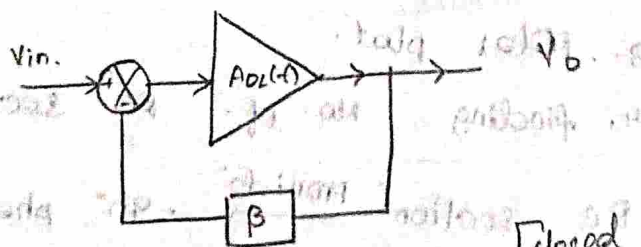
③. stability:

* Open loop circuit op-amp is not used to design amplifier circuit (not advisable)

* due to $A_{OL} = \infty$, when we apply $< 1 \mu V$, the output is amplified give o/p.

But we apply $> 1 \mu V$, the output is amplified ∞ due to gain is ∞ , but in op-amp maximum is supply voltage. so it is saturated, that's why is not preferable.

In comparator circuit we can use open loop op-amp circuits.



[closed loop negative feedback]

gain expression of closed loop operational amplifier.

$$\frac{V_o}{V_{in}} = \frac{A_{ol}(f)}{1 + A_{ol}(f)\beta}$$

The denominator of polynomial is characteristics eqn.

$$1 + A_{ol}(f)\beta = 0$$

$$A_{ol}(f)\beta = -1$$

If $A_{ol}(f)\beta$ is -1, it becomes oscillatory, so we take magnitude.

$$|A_{ol}(f)\beta| = 1$$

* The order of system is ↑sed when considering more no. of capacitor effect.

* If system is higher order system to find stability we can use.

1. Bode plot.
2. Nyquist plot.
3. Polar plot.
4. finding no. of RC section.

1 RC section provides -90° phase shift.

For multiple no. of RC provides multiple no. of phase shift.
 & multiple corner frequency.

⊗ ⇒ Low
 stable system.

* stable
 Bounded

* when

This

⇒ cancel

⇒ use

when m

leads to

Frequency

bm (or)

10m

1. Intern

Frequency

⇒ building

the IC.

[when 10-

operational

① $\Rightarrow$ lower order system is always stable when compared to higher order system.

* stable when Bounded Input & Bounded output.

* when it is linear circuit is stable.

This is attained by.

$\Rightarrow$ cancel out (or) reduce the P_c section.

$\Rightarrow$ use frequency compensation.

when multiple corner frequency $\uparrow$ it leads to unstable system.

Frequency Compensation.

$\Rightarrow$ This reduce the no. of corner frequency exist in circuit.

6m (or)

10m

1. Internal

Frequency compensation.

$\Rightarrow$ building capacitor inside the IC. (built in)

[when 10-30 pF]

2. External frequency Compensation.

we need to combine the op-amp with additional circuit [containing capacitor].

dominant pole.

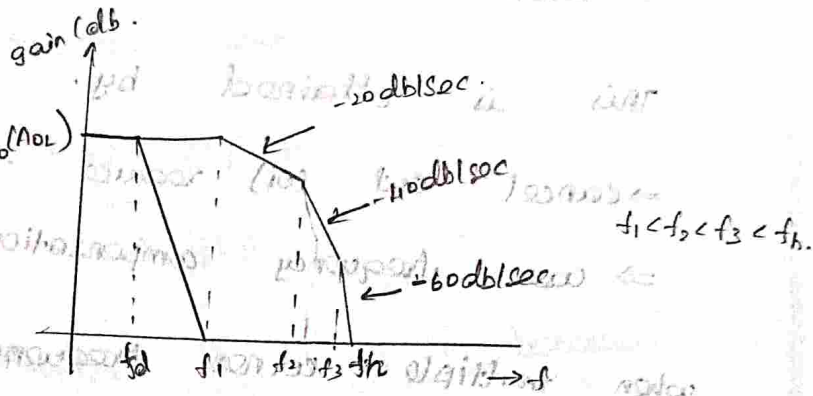
Pole-zero compensation.

dominant - pole. compensation:

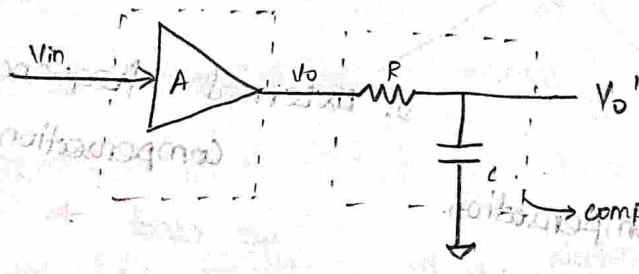


uncompensated circuit

assume 3 corner frequencies



In order to compensate corner frequencies additional circuits is added.



$$\frac{V_o'}{V_{in}} = A \cdot A_1$$

for capacitor C , $X_c = \frac{1}{j\omega C} = -jX_c$;
(impedance)

$$X_c = 1/\omega C.$$

$$V_o' = \frac{V_o (-jX_c)}{R - jX_c} = \frac{-jX_c \cdot V_o}{-jX_c [1 + jRX_c]}$$

$$A_1 = \frac{V_{o1}}{V_o} = \frac{1}{1 + jR\omega C}$$

$$\frac{V_{o1}}{V_o} = \frac{1}{1 + j(f/f_d)}$$

here, $f_d = \frac{1}{2\pi RC}$; $f_d < f_1 < f_2 < f_3 < f_h$

$f_d \rightarrow$ dominant frequency.

I choose proper value of R & C

so the f_d is less than f_1, f_2, f_3

Now $A = \frac{A_{OL}}{[1 + j(f/f_1)][1 + j(f/f_2)][1 + j(f/f_3)]}$

$A_1 \rightarrow$ gain of compensated circuit (R & C)

$$\text{Gain} = A \cdot A_1 = \frac{A_{OL}}{[1 + j(f/f_1)][1 + j(f/f_2)][1 + j(f/f_3)][1 + j(f/f_d)]}$$

(*) $\Rightarrow f_d$ must be less than f_1, f_2, f_3 .

$\Rightarrow f_d$ depends on R & C.

Drawback:

$\Rightarrow$ It is not suitable for application

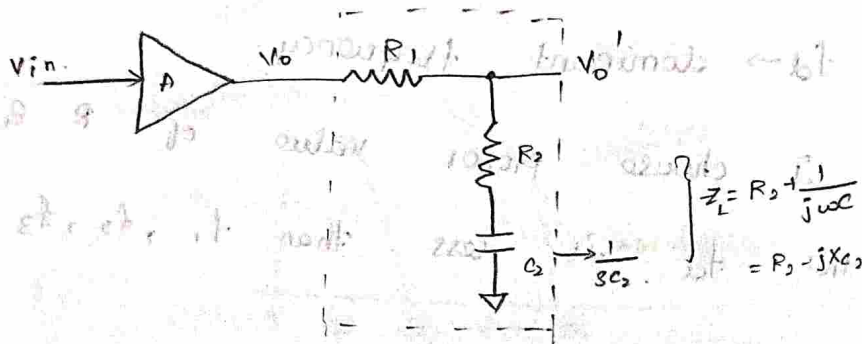
when BW is more.

$\Rightarrow$ It has less bandwidth.

To overcome this we use pole-zero technique.

Here we include poles externally to improve the circuit, that's why is called as dominant - pole.

Pole-zero Compensation:



Here we include zero as well as pole, by which some improvement is done.

$$\frac{V_o'}{V_{in}} = A \cdot A_1$$

$$V_o' = \frac{V_o (R_2 - jX_{C_2})}{R_1 + R_2 - jX_{C_2}}$$

here, $Z_L = R_2 + \frac{1}{sC_2}$

$$Z_L = \frac{1 + sR_2C_2}{sC_2} \rightarrow \textcircled{1}$$

$V_o' \rightarrow$ voltage across Z_L

$$V_{o'} = \left(\frac{V_o}{R_1 + Z_L} \right) \cdot Z_L \rightarrow \textcircled{2}$$

Sub ① in ⑤:

$$V_o' = \frac{V_o \left(\frac{1+SR_2C_2}{sC_2} \right)}{R_1 + \left(\frac{1+SR_2C_2}{sC_2} \right)}$$

$$\frac{V_o'}{V_o} = A_1 = \frac{1+SR_2C_2 / sC_2}{1+SR_1C_2 + SR_2C_2 / sC_2}$$

$$A_1 = \frac{1+SR_2C_2}{1+sC_2(R_1+R_2)}$$

$$A_1 = \frac{1+j\omega R_2C_2}{1+j\omega C_2(R_1+R_2)}$$

$$A_1 = \frac{1+j2\pi f R_2C_2}{1+j2\pi f C_2(R_1+R_2)}$$

$$A_1 = \frac{1+j(f/f_1)}{1+j(f/f_p)}$$

here,

$f_p \rightarrow$ frequency due to insertion of pole.

$$f_p = \frac{1}{2\pi C_2(R_1+R_2)}$$

$$f_1 = \frac{1}{2\pi R_2C_2}$$

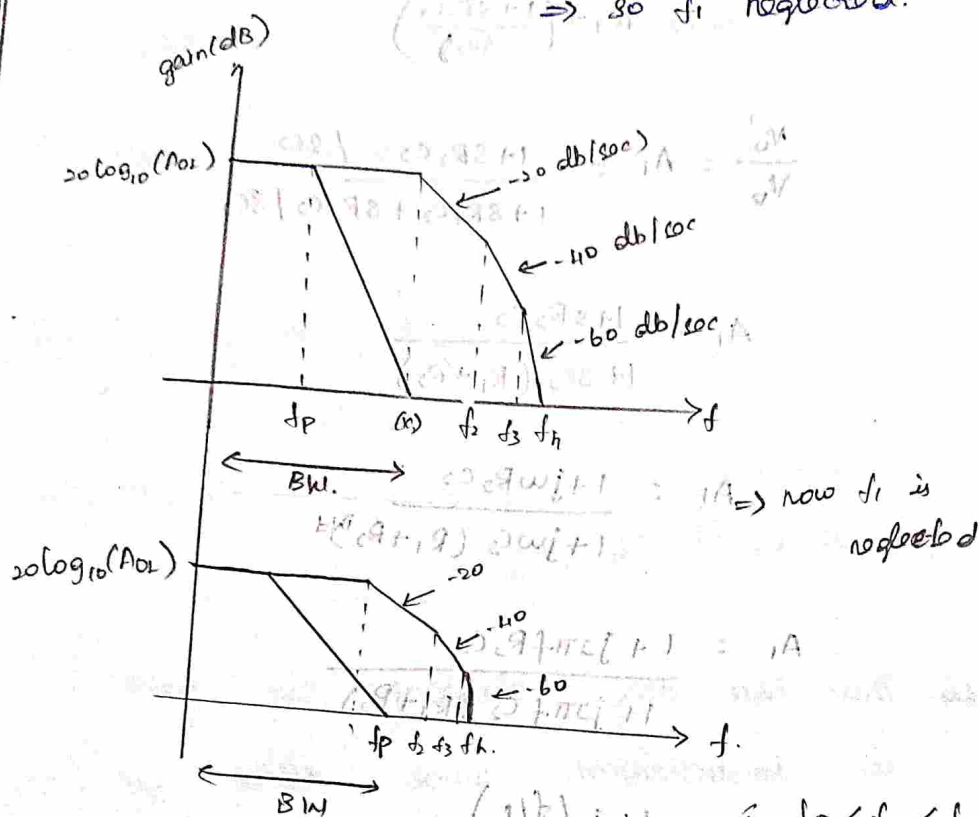
$\Rightarrow$ frequency due to insertion of zero.

To chase R_2 & C_2 [value] so that

the this f_1 & f_p (1st corner frequency) should be same. & it follows.

$$f_1 < f_p < f_2 < f_3 < f_h$$

$\therefore f_1 = f_i$ (due to zero insertion)
 $\Rightarrow$ so f_1 neglected.



$\Rightarrow$ Bandwidth is increased, compare 2 graphs.

overall gain $= A \cdot A_1$

$$\frac{V_o'}{V_{in}} = \frac{A_0 [1 + j(f/f_1)]}{[1 + j(f/f_2)][1 + j(f/f_3)][1 + j(f/f_p)]}$$

$$\frac{V_o'}{V_{in}} = \frac{A_{01}}{[1 + j(f/f_2)][1 + j(f/f_3)][1 + j(f/f_p)]}$$

By using this

$\Rightarrow$ improve the bandwidth.

$\Rightarrow$ cancel the already existing frequency

so it reduces multiple corner frequency.

⑤ Given that
 input com
 voltage
 output

Sol:-

CMRR

(common
 Reject

Q. Given that CMRR of an op-amp is 100 dB
 Input common voltage is 15 V, differential
 voltage gain is 1000. Calculate the
 output common voltage

Sol:

CMRR = $\frac{A_d}{A_c}$ = $\frac{\text{differential mode gain}}{\text{common mode gain}}$
 (Common mode Rejection Ratio)
 $(\text{CMRR})_{\text{dB}} = 20 \log_{10} \left[\frac{A_d}{A_c} \right] = 100 \text{ dB.}$

$$A_d = \frac{V_o}{V_d} = 1000$$

$$A_c = \frac{V_o}{V_c} ; V_c = 15 \text{ V.}$$

$$20 \log_{10} \left[\frac{A_d}{A_c} \right] = 100.$$

$$\log_{10} \left[\frac{A_d}{A_c} \right] = \frac{100}{20} = 5.$$

$$\log_{10} \left[\frac{1000}{A_c} \right] = 5.$$

$$\frac{1000}{A_c} = 10^5$$

$$A_c = \frac{1000}{10^5}$$

$$A_c = 0.01$$

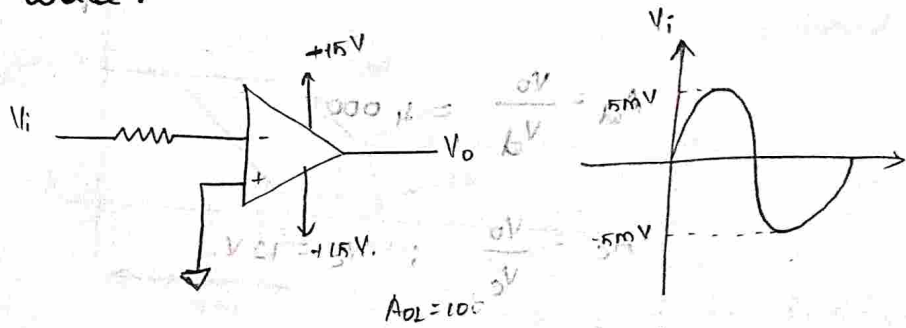
$$A_c = \frac{V_o}{V_c}$$

$$0.04 = \frac{V_o}{12}$$

$$V_o = 0.04 \times 12$$

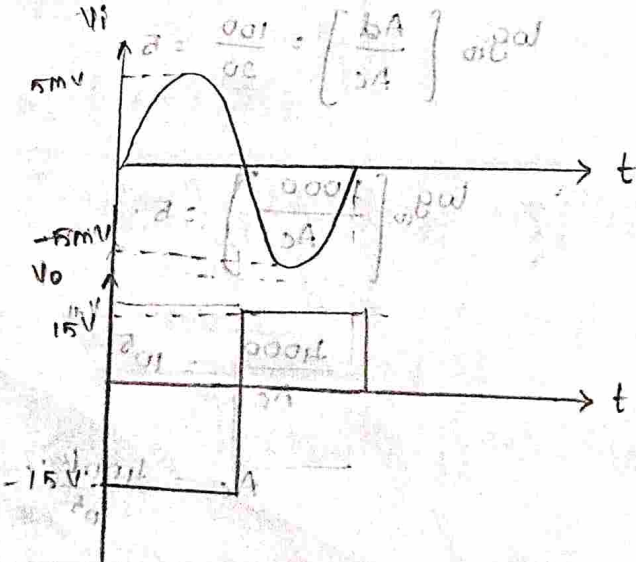
$$V_o = 0.48 \text{ Volt}$$

⑥. Examine the output voltage for the given circuit. Assume that the input V_i is 5mV sine wave.



Sol:

For ideal open loop op-amp, gain is ∞ .



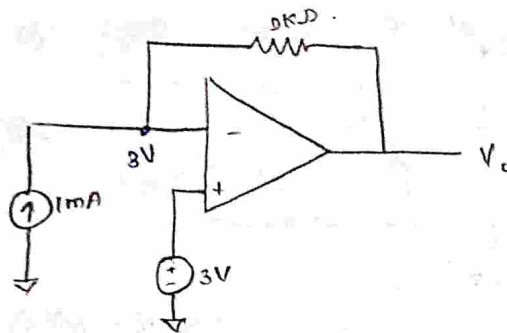
It is called squarewave converter

Assume $A_{OL} = 10^6$, $V_o = A_{OL} \cdot V_{in}$

$$= 5000 > 15 \text{ V.}$$

$$= -5000 < -15 \text{ V.}$$

7. Find the o/p for the given circuit.



2 steps to consider:

1. There is no current flow into the i/p terminal of opamp
2. Use virtual ground concept.

Sol:
Nodal analysis:

$$-1 \times 10^{-3} + \frac{3 - V_o}{2 \times 10^3} = 0$$

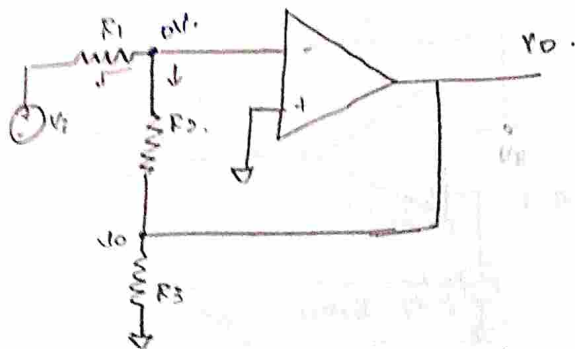
$$\frac{3 - V_o}{2 \times 10^3} = 1 \times 10^{-3}$$

$$3 - V_o = 2 \times 1 \times 10^{-3} \times 10^3$$

$$3 - 2 = V_o$$

$$\boxed{V_o = 1V}$$

8. Assume the op-amp to be ideal. Find the voltage gain of the amplifier shown below



- ⑧ An op-amp has an input offset voltage of 5mV , the closed loop gain is 100 . Find the output offset voltage.

Sol:

$$\frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_2} = 0$$

$$\frac{-V_i}{R_1} = \frac{V_o}{R_2}$$

$$\therefore \frac{V_o}{V_i} = -\frac{R_2}{R_1}$$

[Inverting Amplifier]

Sol:

Given,

$$V_{os} = 5\text{mV}$$

$$A = 100$$

$$V_o = A \cdot V_{os} = 100 \times 5 \times 10^{-3}$$

$$V_o = 0.5\text{V}$$

$$V_o = 500\text{mV}$$

- ⑩ For an op-amp, the i/p offset voltage changes by $30\mu\text{V}$ when the supply voltage

changes
value of
solt

- ⑪ An JFET shows a when the by 5V constant.

Sol:

- i).
ii).

changes from $\pm 15V$ to $\pm 14V$. Find the value of PSRR and $(PSRR)_{dB}$.

Sol:

Given,

$$PSRR = \frac{\Delta V_{os}}{\Delta V_s} \quad ; \quad SVRR = \frac{\Delta V_o}{\Delta V_s}$$

$$\Delta V_{os} = 30 \mu V$$

$$\Delta V_s = 15V - 14V = 1V$$

$$PSRR = \frac{30 \times 10^{-6} V}{1 V} = 30 \mu V \rightarrow \text{amount of error produced}$$

$$(PSRR)_{dB} = 20 \log_{10} \left[\frac{1}{PSRR} \right]$$

$$= 90.457$$

$$\approx 90.46 \text{ dB}$$

- ii) An JFET op-amp configured in open-loop operation shows a change in o/p voltage of $6mV$ when the positive supply voltage changes by $2V$ while the i/p voltage is held constant. i) calculate SVRR in mV/V

ii) convert SVRR into dB.

Sol:

Given,

$$\Delta V_o = 6mV$$

$$\Delta V_s = 2V$$

$$i). \quad SVRR = \frac{\Delta V_o}{\Delta V_s} = \frac{6mV}{2V} = \frac{3mV}{1V}$$

$$ii). \quad (SVRR)_{dB} = 20 \log_{10} \left[\frac{1}{3 \times 10^{-3}} \right]$$

$$\approx 50.46 \text{ dB}$$